Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

Constant Velocity Lab

Criterion C — Processing and evaluating • formative

Newton's idea, tested

In the Newton's Laws lesson, the skydiver showed you the key idea of Newton's First and Second Laws: **when the forces on something are balanced, its speed stays constant.** Today you test that idea on a robot.

Once the robot is moving, the push from its wheels balances the friction against it. So Newton's laws make a prediction you can check with a stopwatch:

> **Prediction:** the forces on the moving robot are **balanced**, so it keeps a **constant speed**. It travels **equal distances in equal times**, and its distance–time graph is a **straight line**.

Constant velocity means a steady speed in a straight line — the "balanced" row of the Newton's First Law table. You will measure that speed with the formula from the speed lesson: speed = distance ÷ time.

The method

One robot for the whole class

Kit: the robot; a 3 m track taped on the floor, with a line every 0.5 m from the start line (0 m) to 3.0 m; six stopwatches.

Jobs for each run

- **Driver** — switches the robot on and sets it down 30 cm behind the start line, pointing straight along the track. The run-up means it is already at its steady speed when timing begins.
- **Starter** — stands at the start line and calls **"Go!"** the instant the front of the robot crosses it.
- **Six timers** — one at each line from 0.5 m to 3.0 m. Start your stopwatch on "Go!" and stop it the instant the front of the robot crosses *your* line.
- **Scribe** — writes each timer's time on the board, to 1 decimal place.

Steps

1. Run the robot. Each timer reads out their time and the scribe writes it in the **Run 1** column on the board.
2. Swap jobs and run it again (**Run 2**), then once more (**Run 3**), so everyone times at least once.
3. Copy all eighteen times from the board into your own table.

Check: each time on the board is bigger than the time at the line before it — the robot reaches the far lines later.

Care: stand beside the track, outside the tape, so the robot's path stays clear.

C1 Your results

The **0 m row is filled in for you**: every stopwatch starts as the robot crosses the start line. Work out each **average** — add the three times and divide by 3 — and round it to 1 decimal place.

Distance / m	Run 1 time / s	Run 2 time / s	Run 3 time / s	Average time / s
0	0	0	0	**0**
0.5				
1.0				
1.5				
2.0				
2.5				
3.0				

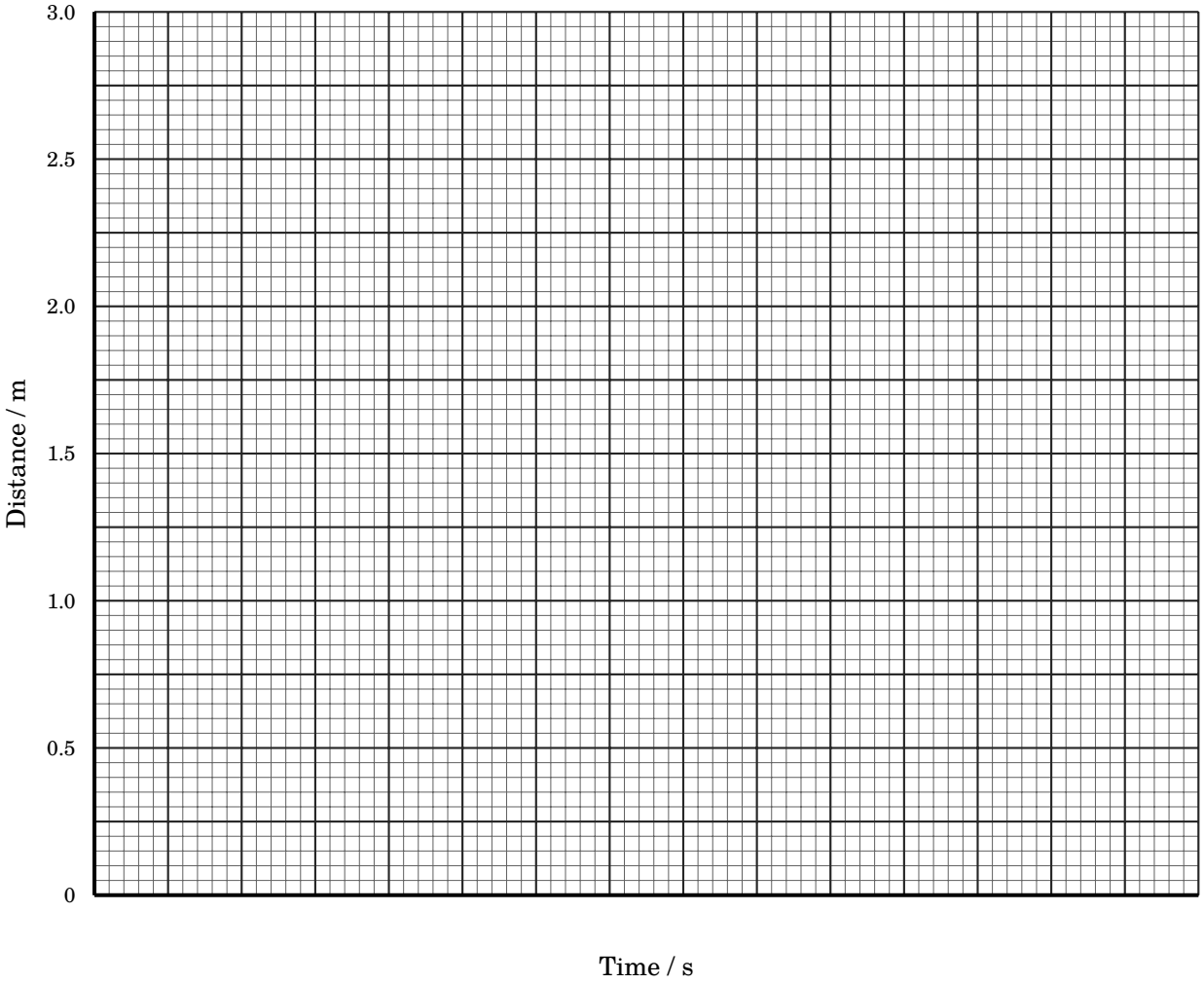
Check: each average lies between the smallest and the largest of its three times.

C1 Your graph

A **distance–time graph**: distance up the side, time along the bottom. The distance axis is numbered for you.

1. **Number the time axis.** Choose **0.5 s, 1 s, 2 s or 5 s** for each big square — the smallest that lets your longest average time fit on the axis.
2. **Plot** each average time against its distance as a small, neat cross. The first point is $(0,\ 0)$.
3. If your points lie close to a straight line, draw **one straight line of best fit with a ruler**. If they curve, draw **one smooth curve**.

Check: your line passes close to every point, with about as many points above it as below.



C2 What your graph shows

Circle the words that match your graph, and fill in the blanks.

Choose the words that fit your graph

My graph is a **straight line** / **curve**, so the robot travelled _____ distances in _____ times.

This **supports** / **does not support** the prediction that the forces on the moving robot are _____.

Back it up with **two numbers from your own table**.

Now work out the robot's **speed** over the whole track, using your average time at 3.0 m. Give it to 2 decimal places.

speed = distance ÷ time = 3.0 m ÷ _____ s = _____ m/s

Check: speed × time gives you back 3.0 m.

C4–C5 How good were your times?

Write it like this

The hardest thing to time exactly was _____.

To make the times more accurate, I would _____.

To go further — equal distances in equal times?

Use your **average** times. For each 0.5 m stretch of track, subtract the time at the start of the stretch from the time at the end of it. Then work out the robot's speed on each stretch: 0.5 m ÷ time, to 2 decimal places.

Stretch / m	0–0.5	0.5–1.0	1.0–1.5	1.5–2.0	2.0–2.5	2.5–3.0
Time / s						
Speed / m/s						

What do you notice about the six speeds, and how do they compare with your speed for the whole track?